

Multilevel modelling



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Introduction

Method with several names

- ▶ multilevel models
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Multilevel

- ▶ Hierarchical / multilevel models because they allow data analysis at several levels of aggregation.
- ▶ E.g. level 1: single data points of a student, level 2: student, level 3: school, level 4: city, level 5: country...
- ▶ The levels are numbered from bottom to top, and from specific to general.
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Why multilevel?

Multilevel data structures

- ▶ Generalization of the simple linear regression and ANOVA to complexer data with nested groups
 - ▶ When hierarchical data is analysed only at one level, this can lead to under- or overestimation of effects, which undermines the interpretability of the findings.
- Ecological and atomistic fallacy, Simpson's paradox

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Ecological fallacy

An ecological fallacy is a fallacy in the interpretation of statistical data. It occurs when a data pattern that was observed at the group level (e.g. the group mean) is generalised to the level of individual data points. This is a fallacy because the pattern at one level of aggregation does not necessarily coincide with the pattern at another level of aggregation.

Ecological fallacy in which the direction of an effect switches between the group and the individual levels: [Simpson's Paradoxon](#).

([Eid et al., 2010](#); [Robinson, 1950](#)).

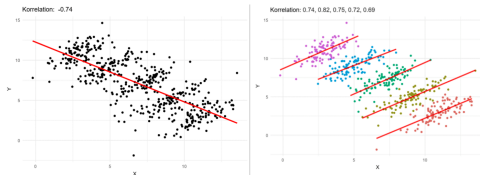
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Ecological fallacy: example



A research team studies the relation between autonomy at the work place and work satisfaction. It conducts interviews in 5 companies, correlates autonomy with satisfaction and finds an overall negative correlation between the two. But the same correlation was positive in each individual company. They conclude that autonomy correlates positively with satisfaction in general, but that other variables such as income also influence satisfaction scores.

Heteroscedasticity

- ▶ Hierarchical data violate the assumption of homoscedasticity (i.e. variance homogeneity) from linear regression.
- ▶ Variance within groups tends to be smaller than variance between groups. (e.g. learn environment of students in the same class is more similar than of students in different classes).
- ▶ In multilevel analyses this is no problem. On the contrary: differences between variances can themselves become the subject of new research hypotheses.

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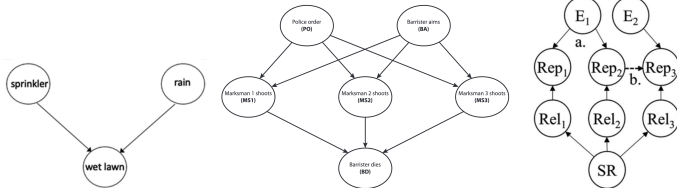
Advantages for theory development

- ▶ Development of more differentiated, abstract theories
- ▶ Models in which one latent variable can be linked to several observations, and one observation to several latent variables (e.g. common-cause, common-effect networks).
- ▶ Modelling of (non)linear time series data

([Lee, 2011](#)).

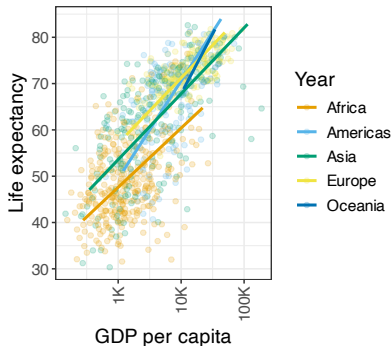
Example for complexer models

Causal networks



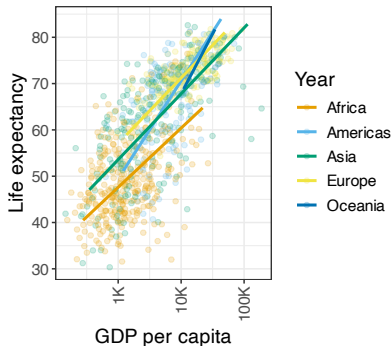
(Lagnado et al., 2013; Sloman & Lagnado, 2015; Pilditch et al., 2025).

Example questions for multilevel data



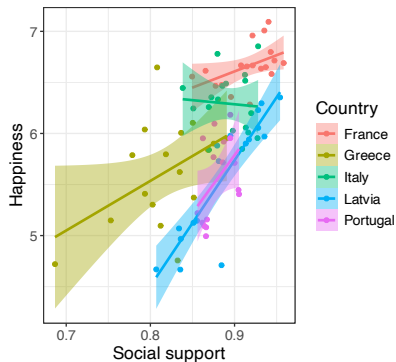
- What is the overall intercept?
- What is the overall regression slope?
- Do the intercepts differ between groups?
- Do the slopes differ between groups?
- Are there cross-level interactions?

Example questions for multilevel data



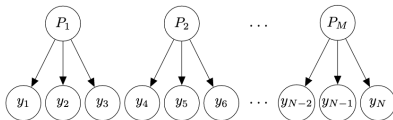
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Example hypothesis about variance

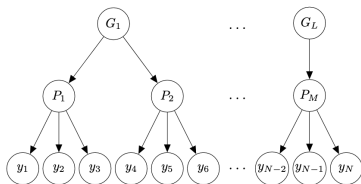


- For the data for the [World Happiness Report 2024](#): Is the variance in people's happiness smaller (or larger) for higher (or lower) values of social support?

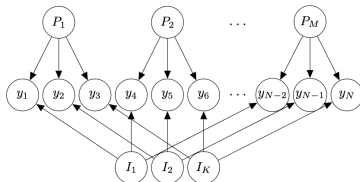
Two or more levels



(a) Single Random Effect

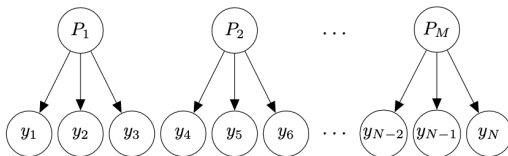


(c) Nested Random Effects



(b) Crossed Random Effects

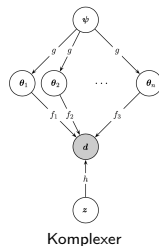
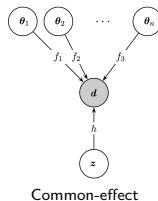
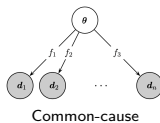
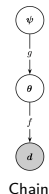
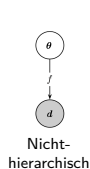
Special case: within-participant designs



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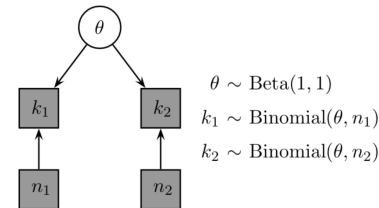
Notation

Notation for causal models

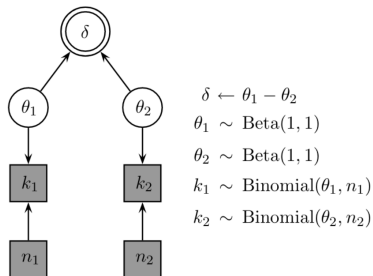


(Lee, 2011).

General graphical notation



θ = gemeinsame Wahrscheinlichkeit,
dass zwei Münzen auf Zahl landen



δ = Differenz zwischen
zwei Wahrscheinlichkeiten

- ▶ **squared**: discrete vs. **round**: continuous
- ▶ **shaded**: observed vs. **unshaded**: latent
- ▶ **double border**: variable value is fully determined by the values of the variables that link to it.

(Lee & Wagenmakers, 2013).

Mathematical notation

For regression-type models there is a variety of conventions;
example for intercept-only model:

Level 1

$$y_{ij} \sim \text{Normal}(\mu_{ij}, \sigma^2)$$

$$\mu_{ij} = \beta_{0i} + x_{ij1}\beta_{1i} + x_{ij2}\beta_{2i} + x_{ij1}x_{ij2}\beta_{3i}$$

Level 2

$$\beta_{0i} \sim \text{Normal}(\gamma_0, \tau_0^2)$$

(Haaf, 2025; [Veenman et al., 2023](#)).

Notation in brms

1. `m1 <- brm(y ~ x1 + x2, data = dat)`
2. `m1 <- brm(y ~ x1 + x2 + (1 | id), data = dat)`
3. `m1 <- brm(y ~ x1 + x2 + (1 | classroom / id),
data = dat)`
4. `m1 <- brm(y ~ x1 + x2 + (1 | classroom) +
(1 | classroom:id), data = dat)`
5. `m1 <- brm(y ~ x1 + x2 + (1 | id) +
(1 | materials), data = dat)`
6. `m1 <- brm(y ~ x1 + x2 + (x1 || id),
data = dat)`

(Bürkner, 2017, 2018).

Notation in `brms` (2)

1. One level
2. Two levels, grouping level: `id`
3. Three levels, trials are nested in individuals (`id`) and individuals are nested in classrooms
4. Equivalent to previous example
5. Two crossed (instead of nested) levels: trials are nested in individuals and in experimental materials
6. Double stroke "`||`": Variances, but no covariances are estimated.

(Bürkner, 2017, 2018).

Which sample size?

- ▶ Data analysed with multilevel models do not have a single sample size, but separate sample sizes at each level of aggregation (e.g. study with 20 trials per participant, 35 participants per school, 10 schools per country).
- ▶ These different sample sizes make it more difficult to determine degrees of freedom for statistical tests, and there is no single agreed on method for doing so.
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How many levels?

Different criteria have been proposed for when to model an additional level of aggregation:

- ▶ When there are 5 or more data points per cell in the design
- ▶ When there are several measurements per individual (within design).
- ▶ When individuals are in groups (e.g. classrooms in schools in districts; doctors in hospitals; plants in fields)
- ▶ When we want to generalise beyond the specific individuals / experimental conditions measured.
- ▶ When we want to estimate not only means but also variances.
- ▶ When the variance at the group level makes up a certain percentage (e.g. 10% of the total sample variance).

(Gelman, 2005).

Interclass correlation

- ▶ The intraclass correlation (ICC) is the proportion of level 2 variance out of the total variance.
- ▶ For trials nested in individuals: proportion of the total variance attributable to interindividual differences.

$$ICC = \frac{\sigma_{level2}^2}{\sigma_{level1}^2 + \sigma_{level2}^2}$$

Rule of thumb: open an aggregation level when $ICC \geq 10\%$.

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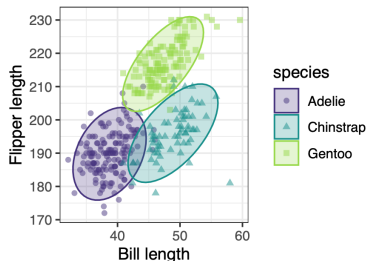
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Examples with brms

Penguin research



```
1 > head(penguins)
2
3   species    island bill_len bill_dep flipper_len body_mass
4 1  Adelie Torgersen   39.1    18.7         181      3750
5 2  Adelie Torgersen   39.5    17.4         186      3800
6 3  Adelie Torgersen   40.3    18.0         195      3250
```

Can flipper length be predicted from bill length? To what extent does this vary between species, and/or between islands?

From hypotheses to models

```
1 m1 <- brm(flipper_len ~ bill_len
2           , data = penguins
3           , warmup = 1000
4           , iter = 10000
5           , chains = 4
6           , cores = 4
7           , prior = prior_2
8           , sample_prior=TRUE
9           , control = list(adapt_delta = 0.99,
10                          max_treedepth = 12))
# , save_pars = save_pars(all = TRUE)
```

```
1 m2 <- brm(flipper_len ~ bill_len + (1 | species))
```

```
1 m3 <- brm(flipper_len ~ bill_len + (1 | island))
```

```
1 m4 <- brm(flipper_len ~ bill_len + (1 | species) + (1 |
  island))
```

Model comparisons

```
1 # Forwards selection
2 bf_10 <- bayes_factor(m1, m0) # 1 better than 0, BF > 100
3 bf_21 <- bayes_factor(m2, m1) # 2 better than 1, BF > 100
4 bf_31 <- bayes_factor(m3, m1) # 3 better than 1, BF > 100
5 bf_42 <- bayes_factor(m4, m2) # 2 better than 4, BF ~ 5
6 bf_43 <- bayes_factor(m4, m3) # 4 better than 3, BF > 100
```

- ▶ Model *m2*, with a random intercept for species but not for island, accounted best for the pattern in the data.
- ▶ This comparison was suboptimal because the group level variable (species) has only three values. Rule of thumb: minimum 5 values, to be able to compute their variance.

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Model *m2* summary

```
> summary(m2)
Family: gaussian
Links: mu = identity
Formula: flipper_len ~ bill_len + (1 | species)
Data: penguins (Number of observations: 342)
Draws: 4 chains, each with iter = 10000; warmup = 1000; thin = 1;
       total post-warmup draws = 36000

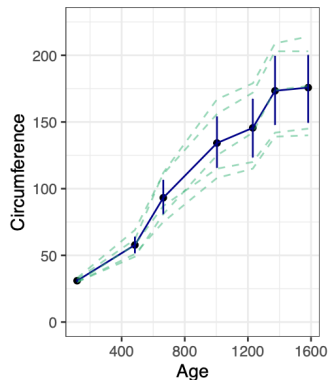
Multilevel Hyperparameters:
~species (Number of levels: 3)
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
sd(Intercept)    15.29     7.89    6.39   35.42 1.00   10611
      Tail_ESS
sd(Intercept)   15268

Regression Coefficients:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept    151.58     9.07   133.29   169.45 1.00   12469   14048
bill_len       1.08     0.11     0.87     1.29 1.00   22161   19174

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma       5.85     0.23     5.42     6.31 1.00   21471   18131
```

Orange trees

```
1 > Orange <- data.frame(  
2   Orange)  
3  
4   Tree  age  circumference  
5 1     1  118             30  
6 2     1  484             58  
7 3     1  664             87  
8 4     1 1004            115
```



Do the stems of orange trees grow over time?

Treegrowth - one level

```
1 > m0 <- brm(circumference ~ age
2           , data = Orange
3           , sample_prior = TRUE
4           , warmup = 1000
5           , iter = 10000
6           , save_pars = save_pars(all = TRUE)
7           )
8 > summary(m0)
```

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$$\beta_0 = 17.43, \beta_1 = 0.11, \sigma = 24.62.$$

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	17.43	9.03	-0.42	35.44	1.00	30887	23870
age	0.11	0.01	0.09	0.12	1.00	30925	24256

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	24.62	3.17	19.31	31.78	1.00	29000	24687

Treegrowth - within-design

```
1 > m1 <- brm(circumference ~ age + (1|Tree)
2   , data = Orange
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```

$$\beta_0 = 17.35, \beta_1 = 0.11, \sigma = 15.97, \tau_0 = 28.29.$$

Multilevel Hyperparameters:

~Tree (Number of levels: 5)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	28.29	14.61	11.54	66.54	1.00	7457	11710

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	17.35	14.76	-12.33	46.35	1.00	8598	10789
age	0.11	0.01	0.10	0.12	1.00	24329	20984

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	15.97	2.23	12.34	21.08	1.00	17081	18291

Treegrowth - model comparison



```
1 > bayes_factor(m1, m0)
2
3 Estimated Bayes factor in favor of m1 over m0:
   1993.45591
```

- ▶ Model `m1` with two levels (single measurements on level 1, individual trees on level 2) accounts much better for the data than model `m0` with just one level.
- ▶ There is thus evidence for individual differences between the trees.

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